

DL Graded Quiz M5

Question 1

In Hidden Markov models, we can combine forward and backward recursion to compute $P(\mathbf{x}_j | \mathbf{x}_{-j})$. But what does this probability represent?

- ☐ The probability of observation $\mathbf{x}_j$ taking into account all available information but that at point j
- ☐ The probability that observation j is less than a given threshold
- ☐ Correlation between $\mathbf{x}_j$ and $\mathbf{x}_{-j}$
- ☒ The joint probability of forward and backward recursion

Question 2

Suppose we decide to split our sample to devote 70% of it to training. The training set of inputs (X) in our LSTM model has a shape of (16800, 25, 5). How many different input (or series of inputs) are we considering?

- ☐ 16800
- ☐ 11760
- ☐ 25
- ☒ 5

Question 3

Suppose in an LSTM the input gate value is equal to zero, while that of the forget gate is equal to 1. Then, the value of the memory cell C_t will do which of the following?

- ☐ Be close to zero
- ☐ Resemble the previous value C_{t-1}
- ☐ Be close to one
- ☒ Resemble the candidate cell $\tilde{C}_t$

Question 4

Select which of the following statements is correct:

- ☐ Bidirectional LSTM models are not always more appropriate, but they deliver better performance than unidirectional ones.
- ☐ There are some tasks for which bidirectional models are simply not appropriate.
- ☐ Bidirectional LSTM models are always more appropriate and deliver better performance than unidirectional ones.
- ☒ Unidirectional LSTM models are always more appropriate and deliver better performance than bidirectional ones.

Question 5

If the forward hidden state of a bidirectional LSTM contains 84 units, how many units will the backward hidden state contain?

- ☐ 42
- ☐ 6
- ☐ 36
- ☒ 84

Question 6

In GRUs, the candidate hidden state is determined using which of the following?

- ☐ tanh activation function
- ☐ sigmoid activation function
- ☐ Input gate
- ☒ Forget gate

Question 7

Why do we use sigmoid activation functions for Reset and Update gates in GRUs?

- ☐ We want positive output in the $(0, \infty)$ interval.
- ☐ We want output in the $(-1, 1)$ interval.
- ☐ It is the only one capable of dealing with large amounts of data.
- ☒ We want output in the $(0, 1)$ interval.

Question 8

Suppose you have a model with bias terms and 1 bidirectional LSTM layer consisting of 56 units. This layer takes inputs from 250 units. How many more parameters will be trained relative to the unidirectional case?

- ☐ 14000
- ☐ 137536
- ☐ 68768
- ☒ 0

Question 9

If the forward hidden state of a bidirectional LSTM contains 24 units, how many units will the backward hidden state contain?

- ☐ 12

- ☐ 24
- ☐ 6
- ☒ 48

Question 10

Suppose you have a model with bias terms and 1 LSTM layer consisting of 125 units. This layer takes inputs from 65 units. How many parameters will be trained at the forget gate ?

- ☐ 90500
- ☐ 22850
- ☐ 95500
- ☒ 23875

Question 11

How can we define a candidate memory cell, $\tilde{C}_t$?

- ☐ .
- ☐ $\tilde{C}_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t$
- ☐ $\tilde{C}_t = F_t \odot C_{t-1} - I_t \odot \tilde{C}_t$
- ☒ $\tilde{C}_t = \tanh(X_t W_{xc} + H_{t-1} W_{hc} + b_c)$

Question 12

Suppose you have a model with bias terms and 1 bidirectional LSTM layer consisting of 256 units. This layer takes inputs from 15 units. How many more parameters will be trained relative to the unidirectional case?

- ☐ 557056
- ☐ 278528
- ☐ 0



Time left:
1:29:33

QUESTIONS

- | | | | |
|----|----|----|----|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Question 13

The probabilistic graphical model that we use to illustrate bidirectional RNNs is which of the following?

- ☐ A Hidden Markov Model
- ☐ A Monte Carlo model
- ☐ A Hidden Poisson Model
- ☒ A Latent dynamic model

Question 14

Suppose you have a model with bias terms and 1 LSTM layer consisting of 80 units. This layer takes inputs from 65 units. How many parameters will be trained at the output gate?

- ☐ 11680
- ☐ 46720
- ☐ 11600
- ☒ 46400

Question 15

Which activation function do we use to compute the candidate memory cell in LSTMs?

- ☐ sigmoid
- ☐ ReLU
- ☐ tanh
- ☒ linear

Question 16

Suppose you have a model with bias terms and 1 bidirectional LSTM layer consisting of 512 units. This layer

takes inputs from 150 units. How many parameters will be trained in total?

- ☐ 2210630
- ☐ 2715648
- ☐ 1357824
- ☒ 1286320

SUBMIT

